Groups and Rotations

This is a Groups and rotations packet for the students of Physics 736-C: Advanced Studies; Mechanics.

Linear algebra in 3D

In class we learned about vectors as objects that have various properties like addition, a zero, and multiplication by real numbers. When we have a dot product, vectors gain additional properties like lengths and relative angles. In these notes we will work through some material on rotations. To get interesting results, we will work in 3D.

Vectors and Covectors

1. Let $\mathbf{a} = (a^1, a^2, a^3)^T$. The ^{T} stands for "transpose" and means that

$$\mathbf{a} = \begin{pmatrix} a^1 \\ a^2 \\ a^3 \end{pmatrix}$$

   Is a 3-component column vector. Strictly, the qualifier "column" is unnecessary, since–by convention–all vectors are columns. Note that the $^1, ^2, ^3$ are "upstairs" on the components of $\mathbf{a}$. We write these components as a^i with the "index" $i = 1, 2, 3$.

2. Dual to vectors are "covectors". These are row-shaped:

$$\alpha = (\alpha_1, \alpha_2, \alpha_3)$$

   Note that the $_1, _2, _3$ are "downstairs" instead of "upstairs" as on the vector. We write these components of the covector as α_i with the "index" $i = 1, 2, 3$.

3. The covectors are dual to the vectors in the sense that we can combine the covector with the vector thus: This is an example of "matrix multiplication". In this case between a 1×3 matrix called a row matrix and a 3×1 matrix called a column matrix. When using components, we write this as

$$\alpha(\mathbf{a}) = \sum_{i=1}^{3} \alpha_i a^i$$

where $\sum$ (capital sigma) stands for "sum" and instructs you to take the sum of all the terms $\alpha_i a^i$ for $i = 1, 2, 3$. When written in this way, we say that the result is given in "components" or "index notation".

Einstein summation convention

The symbols are cumbersome and unnecessary, especially when multiple contractions are being carried out:

1. When writing $\sum_{i=1}^{n}$ we are already making the index explicit, and there is exactly one sum for each pair of repeated indices.
2. The range is generally understood from context.
3. When an index is repeated, it is almost always going to be summed.
4. If indices are being used correctly, summed pairs always appear with one of them "up" and one of them "down".[1]

Matrices

Vectors and covectors are simple examples of matrices that have one column or row, respectively. In general a matrix can have m rows and n columns:

$$M = \begin{pmatrix} M^1{}_1 & M^1{}_2 & \dots & M^1{}_n \\ M^2{}_1 & M^2{}_2 & \dots & M^2{}_n \\ \vdots & \vdots & \ddots & \vdots \\ M^m{}_1 & M^m{}_2 & \dots & M^m{}_n \end{pmatrix}$$

This is an example of an $m \times n$ matrix. The elements of this matrix are written $M^i{}_j$ for $i = 1, \dots, m$ and $j = 1, \dots, n$. Sometimes matrices can be multiplied by vectors or covectors or among one another.

1. An $m \times n$ matrix $M^i{}_j$ can be multiplied by an $n \times p$ matrix $N^j{}_k$ to form an $m \times p$ matrix $(MN)^i{}_k$. This is done by multiplying each column of N into the rows of M sequentially according to the rule [MatMult]: In index notation

   $$(MN)^i{}_k = \sum_{j=1}^{3} M^i{}_j N^j{}_k \quad \text{where} \quad i = 1, \dots, m \quad \text{and} \quad k = 1, \dots, p$$

   Note that for the sum to make sense, the second index on M must have the same range as the first index on N. After we sum over this index, it disappears and the result does not depend on n.

[1] Physicists and mathematicians often disagree about which type is called "covariant" and which type is called "contravariant". If the group action is defined $V \to V$, it seems natural to call this "covariant". Then the dual action $V* \to V*$ is called contravariant.

2. A special case is when an $n \times n$ matrix A acts on an n-component vector $\mathbf{u}$: Recall that an n-component vector is an $n \times 1$ matrix. Then

$$A\mathbf{u} = \begin{pmatrix} A^1{}_1 & A^1{}_2 & \dots & A^1{}_n \\ A^2{}_1 & A^2{}_2 & \dots & A^2{}_n \\ \vdots & \vdots & \ddots & \vdots \\ A^n{}_1 & A^n{}_2 & \dots & A^n{}_n \end{pmatrix} \begin{pmatrix} u^1 \\ u^2 \\ \vdots \\ u^n \end{pmatrix} = \begin{pmatrix} A^1{}_1 u^1 + A^n{}_2 u^2 + \dots + A^n{}_n u^n \\ A^2{}_1 u^1 + A^2{}_2 u^2 + \dots + A^2{}_n u^n \\ \vdots \\ A^n{}_1 u^1 + A^n{}_2 u^2 + \dots + A^n{}_n u^n \end{pmatrix} = \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix} = \mathbf{v}$$

Rotations

We now use the stuff above to do some problems specific to 2D and 3D.

1. []{#SO(2) label="SO(2)"} Let

$$R_{12}(\alpha) \ := \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}$$

   1. Describe in detail what this matrix does to vectors in $\mathbf{R}^2$.
   2. Calculate $R_{12}(\alpha)R_{12}(\beta)$ and $R_{12}(\beta)R_{12}(\alpha)$. How are they related?
   3. Calculate the inverse $[R_{12}(\alpha)]^{-1}$, the transpose $[R_{12}(\alpha)]^t$ (relative to the Euclidean norm[2]), and $R_{12}(-\alpha)$. How are they related?
   4. Calculate the determinant $\det R_{12}(\alpha)$. Considering what a determinant usually is, is this interesting?
   5. Think about all the different matrices $\{R_{12}(\alpha)\}_\alpha$ as you let α vary.
      1. What type of space is this? Hints: It is not a vector space. What are the distinct values of α?
      2. What is the meaning of L_{12} (defined below) in this interpretation?

2. []{#so(2) label="so(2)"} Define

$$L_{12} \ := \left. \frac{dR_{12}(\alpha)}{d\alpha} \right|_{\alpha=0}$$

   1. Compute what it is explicitly as a matrix.
   2. Define $(L_{12})^0 := \mathbf{1}$ to be the (2×2) unit matrix, and calculate $(L_{12})^n$ for $n \in \mathbf{N}$.
   3. Define the exponential $e^M = \exp(M)$ of a matrix M by

   $$\exp(M) \ := \sum_{n=0}^{\infty} \frac{1}{n!} M^n$$

   Prove/argue that the sum converges, and explain clearly what type of object this is.

[2] If you think you know what a transpose is, but don't understand why I made the parenthetical remark, then just ignore the remark.

   4. Calculate $\exp(\alpha L_{12})$.[3]

   5. Take a moment to appreciate how remarkable this is. L_{12} is only the first non-trivial term in a MacLauren series for $R_{12}(\theta)$ (*i.e.* it is the infinitesimal part of the finite transformation), but it can be exponentiated to recover the entire finite transformation: In this case integration is just exponentiation.

3. Generalize $R_{12}(\alpha)$ to a 3×3 matrix that rotates any vector in $\mathbf{R}^3$ as follows: It leaves the 3rd component alone and rotates the first two as before. More precisely: We pick a linear subspace $\mathbf{R}^2 \subset \mathbf{R}^3$ that we call the 12-plane, and use it to decompose $\mathbf{R}^3 = \mathbf{R}^2 \oplus \mathbf{R}$.

   1. Explicitly define our new matrix $R_{12}(\alpha)$ to act as before in that plane and as the identity in the complement.

   2. Repeat all steps in problem [so(2)] for this new 3×3 matrix.

4. Define two new matrices $R_{23}(\beta)$ and $R_{31}(\gamma)$ by permuting $1 \to 2 \to 3 \to 1$ in the obvious way. That is, decomposing $\mathbf{R}^3 = (\mathbf{R} \oplus \mathbf{R}) \oplus \mathbf{R}$ further and calling the factors 1, 2, and 3, respectively, $R_{23}(\beta)$ should act as a rotation in 2 and 3 and $R_{31}(\gamma)$ as a rotation in 3 and 1.

   1. Construct $R_{23}(\beta)$ and $R_{31}(\gamma)$ explicitly. (If you notice a sign ambiguity, resolve it by thinking of the 3-1 plane as the orientation reversal of the 1-3 plane.) Warning: The intention should be obvious but the matrices will look different.

   2. Comment on whether/how the properties in problem [SO(2)] change for these new matrices.

   3. Repeat all steps in problem [so(2)] for these new matrices.

5. Next, we consider how matrix products of Rs act on vectors.

   1. Explain in words what the product $R_{23}(\beta)R_{12}(\alpha)$ does to a vector in $\mathbf{R}^3$.

   2. Does $R_{12}(\alpha)R_{23}(\beta)$ do the same thing?

   3. What does $R_{31}(\gamma)R_{23}(\beta)R_{12}(\alpha)$ do? In this context, , , and are called Euler angles.

6. Meditate deeply on the set $\{R_{12}(\alpha), R_{23}(\beta), R_{31}(\gamma)\}_{\alpha,\beta\gamma}$. Is there any combination of products of the Rs that is not a rotation in 3D? Can I get any rotation from such a product? Thinking of the space of all possible rotations in $\mathbf{R}^3$, is it a linear space? What is the dimension of the space? Hint: Use your physical intuition about what this is describing to answer these questions. (I'm not asking for proofs here.)

[3] The element L_{12} is called a rotation "generator". If you did this calculation correctly, you will understand this nomenclature because L_{12} generates finite rotations.

7. We call the set of all rotations with these additional properties the $SO(3, \mathbf{R})$ or just $SO(3)$ when the reality is implied. If $R \in SO(3)$ is any rotation, what is true about the transpose R^t? What is $\det(R)$? What would you call the set of 2×2 matrices?

8. If you have managed to do all the problems above, you have shown that the set of all rotations of Euclidean 3-space is a "group". By definition, this means:

    1. It is a closed set under an associative multiplication operation (which here is matrix multiplication).

    2. There is a unit element (the unit matrix).

    3. For every rotation, there is a (unique) inverse rotation.

    You have also shown that this is true for $SO(2)$, however:

9. The "commutator" $[\cdot, \cdot]$ of two matrices A and B is defined as

    $$[A, B] = AB - BA$$

    A group is called "commutative" or "Abelian" if the commutator of any two elements is 0. Otherwise, it is called non-commutative or non-abelian. What are the correct statements about the (non-)commutativity of $SO(2)$ and $SO(3)$?

10. Calculate the commutators of the matrices L_{12}, L_{23}, and L_{31}. What do you notice?

11. We call a linear map $V \to V$ from a vector space V to itself a "linear operator" or just "operator" when linearity is implied. An operator N is called "normal" if it commutes with its transpose, that is $[N, N^t] = 0$. Show that rotations are normal operators.

12. Use the fact that orthogonal operators are normal to deduce the conditions on the generators corresponding to the following conditions on operators:

    1. The determinant $= 1$

    2. The operator is orthogonal

    3. Use these results to calculate the dimension of the group $SO(n)$ for any $n \in \mathbf{N}$.

13. Up until now, we implicitly assumed we were working in Euclidean space, that is, a(n affine,) real vector space of dimension 3 equipped with a specific inner product[4] called the dot product $\langle u, v \rangle = uv$. Specifically, denoting the components of u and v by $u^{i}{}_{i=1}^{3}$ and $v^{i}{}_{i=1}^{3}$, respectively, the dot product

[4] In more modern language, the inner product is a symmetric bilinear form on the vector space $V = \mathbf{R}^3$. A linear form is a map $V \to \mathbf{R}$ from the vector space V to the field over which it is defined ($\mathbf{R}$ in this case). In a local set of coordinates $x^{i}{}_{i=1}^{3}$ for $\mathbf{R}^3$, the basis vectors are denoted $\partial/\partial x^{i}{}_{i=1}^{3}$ (which we will abbreviate as $\partial_i{}_{i=1}^{3}$) and the linear forms are denoted $dx^{i}{}_{i=1}^{3}$.

is defined as $uv = \sum_{i=1}^{3} u^i v^i$.
A vector space V equipped with an inner product $\langle \cdot, \cdot \rangle$ is called an inner product space. In such a space, for any operator $L : V \to V$, we can define an operator $L* : V \to V$ by setting

$$\langle L * u, v \rangle = \langle u, Lv \rangle \quad \forall u, v \in V$$

This $L*$ is called the adjoint of L w.r.t. to the inner product $\langle \cdot, \cdot \rangle$ of V. This is almost always abbreviated to "the adjoint (of L)".

1. Find the adjoints of the transformations you worked out above.

2. By extending all of this to $\mathbf{R}^n$, find the largest group of linear transformations that preserves the Kronecker product.

Synthesis

In this section, we studied rotation groups of $\mathbf{R}^n$ by looking at some low-dimensional examples. These groups are subgroups[5] of a larger groups of linear transformations of $\mathbf{R}^n$. The largest is called the general linear group of $\mathbf{R}^n$; it is denoted $GL(\mathbf{R}^n)$ or $GL(n, \mathbf{R})$.[6] Since group transformations must have inverses, the only condition on such a transformation (besides linearity) is that it be invertible. Equivalently, the determinant of such a transformation is not allowed to vanish. Since this is an "open" condition, the dimension of $GL(\mathbf{R}^n)$ is n^2.

There were a few main subgroups of $GL(\mathbf{R}^n)$ that we considered:

- The subgroup $SL(\mathbf{R}^n) \subset GL(\mathbf{R}^n)$ of transformations with determinant 1 is called the "special" group. It has dimension $n^2 - 1$.

- The subgroup $O(n) \subset GL(\mathbf{R}^n)$ of matrices that preserve the Kronecker product is called the "orthogonal" group. From your work on the exponential, you know the dimension of this group should be the same as that of the anti-symmetric $n \times n$ matrices, so $\frac{n(n-1)}{2}$.

  Exercise 1. *Check this.*

In this language, the inner product g is denoted

$$g = \sum_{i,j=1}^{3} \delta_{ij} dx^i \otimes dx^j$$

Here δ_{ij} is an array called the "Kronecker delta". It is defined to have value 1 when $i = j$ and 0 otherwise.
Suggested exercise: Prove that if a bilinear form is symmetric, constant, and positive definite, then there is a choice of coordinates that reduces it to the Kronecker product. (Positive definite means $\langle u, u \rangle > 0 \ \forall u \in V$.)

[5] A subgroup $H \subset G$ of a group G is a subset of G that is also a group.

[6] In general, if V is a vector space, the group consisting of all linear transformations $V \to V$ is denoted $GL(V)$. The notation $GL(n, \mathbf{F})$ is used for the group of $n \times n$ matrices with entries valued in the field $\mathbf{F}$. For us, $\mathbf{F} = \mathbf{R}$ or $\mathbf{C}$ (most of the time).

- The group $SO(n) = O(n) \bigcap SL(\mathbf{R}^n)$ is called the "special orthogonal group".

  Exercise 2. *What's an example of an orthogonal transformation that is not special?*

  Exercise 3. *Check that his group also has dimension $\frac{n(n-1)}{2}$.*

The groups G above are all interesting and very nice[7] spaces called smooth manifolds. (This is in addition to being groups.) On such spaces, you can do calculus, and in particular, you can define (co)tangent spaces. As a group one point is special, because it corresponds to the identity transformation $\mathbf{1}$; its tangent space is denoted $T_{\mathbf{1}}G$ and pronounced "the tangent space to G at the identity". It is traditional to denote $T_{\mathbf{1}}G$ by the frakture letter(s) corresponding to the lowercase of whatever G is, so $\mathfrak{g}$ in this case. From general considerations, this is a real vector space with the same dimension as G. The $\mathbf{0} \in \mathfrak{g}$ origin of the vector space corresponds to the point $\mathbf{1} \in G$.

For $G = SO(2)$ and $SO(3)$, you computed a basis of $T_{\mathbf{1}}G$ showing that $\mathfrak{so}(2) \simeq \mathbf{R}$ and $\mathfrak{so}(3) \simeq \mathbf{R}^3$. Since we are talking about matrices, the tangent space to the group at the identity is actually an associative algebra, but this is not necessarily the case in general. There is something that generalizes and makes $\mathfrak{g}$ into a non-associative algebra called the Lie algebra of G, and it goes as follows: Since we were talking about matrix groups, the commutator is defined. The commutator has an explicit expression in terms of the associative matrix multiplication, but it also has the properties that

- it is an alternating (*i.e.* antisymmtric) bilinear map $\mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$.
- it satisfies the Jacobi identity: for any $a, b, c \in \mathfrak{g}$

$$[[a, b], c] + [[b, c], a] + [[c, a], b] = 0$$

Exercise 4. *Show that the commutator satisfies these identities. In particular, show that it satisfies the Jacobi identity precisely because the matrix multiplication is associative.*

We can think of these identities as conditions generalizing associativity of a product in an algebra. Because of this, the combination $[a, b, c] = [[a, b], c] + [[b, c], a] + [[c, a], b]$ is sometimes called the "associator" (but "Jacobiator" is more common).[8]

[7] When this term is used in the technical sense, it means that something has a lot of additional structures on it that allow you to do things you cannot assume you may do in general. Good examples are whether you can put a distance measure on it (metric structure) and/or whether you are able to define derivatives (differentiable structure).

[8] This is probably TMI, but $\mu_2(a, b) = [a, b]$ and $\mu_3(a, b, c) = [a, b, c]$ are the first two non-trivial examples of an infinite collection of "higher products" required to satisfy a coherence condition (similarly to how μ_3 is defined in terms of two μ_2s). This is the structure of a beast called an "infinity algebra (of Lie type)" or L_∞-algebra. Finding simple examples of such algebras is an active area of current research (and potentially an avenue for student research).

In general, any vector space $\mathfrak{g}$ over a field $\mathbf{F}$ with a binary operation $[\bullet, \bullet] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ satisfying the rules above is called a "Lie (pronounced"lee") algebra". The alternating form is called the Lie bracket. Note that the conditions above do not imply anything about how you can define the Lie bracket explicitly. In particular there does not need to be any way to associate a "product" ab to elements $a, b \in \mathfrak{g}$.[9]

Exercise 5. *Show that the cross product* $\times$ *in* $\mathbf{R}^3$ *makes* $(\mathbf{R}^3, \times)$ *into a Lie algebra. Which Lie algebra is it?*

The exponential map $M \mapsto e^M$ was defined on arbitrary matrices, but with a bit more differential geometry, we can define it more generally. In particular, there is a map

$$\exp : \mathfrak{g} \to G$$

called the exponential map that agrees with your calculations in the special case that $\mathfrak{g}$ is an associative algebra. It is important to understand that this map need be neither injective nor surjective.

Exercise 6. *Use your answer to exercise 2 to come up with an example where surjectivity fails.*

It turns out the if the group is compact,[10] then this is the only way it can fail: If G is connected and compact, then the exponential map is surjective.

Exercise 7. *Consider the familiar exponential map* $x \mapsto e^x$ *from algebra and show that it is injective but not surjective. In light of the previous claim, why does it fail to be surjective? Generalize to the map to the complex domain and then consider the image of* $\mathbf{R}$ *in* $\mathbf{C}$ *defined by* $x \mapsto e^{ix}$*. Show that this map is not injective. In light of the previous claim, why doesn't it fail to be surjective?*

Since the "linearization map" $G \to \mathfrak{g}$ is the local inverse to $\exp : \mathfrak{g} \to G$, and since linearization is differentiation, we see that "integration" corresponds to exponentiation. This is a remarkable thing, because in calculus the first derivative of a function f is not usually even close to enough to reconstruct the whole function.

Exercise 8. *With this observation in mind, suppose you have two Lie groups* G *and* H *with isomorphic algebras* $\mathfrak{g} \simeq \mathfrak{h}$*. What will be true about their Lie groups* G *and* H*?*

[9] It's quite the pain in the *derriere* that we cannot assume $\mathfrak{g}$ is associative. However, thanks to a theorem by Poincaré, Birkhoff, and Witt, any Lie algebra $\mathfrak{g}$ can be embedded into an associative algebra $U(\mathfrak{g})$ with identity in such a way that the Lie bracket in $\mathfrak{g}$ becomes the commutator in $U(\mathfrak{g})$. This algebra is called "the universal enveloping algebra of $\mathfrak{g}$".

[10] The formal definition of compact requires point set topology, but there is an equivalent definition for spaces we can embed into $\mathbf{R}^n$ for some n large enough (c.f. Whitney Embedding Theorem). In this case the notion of compactness lines up with intuition: $G \subset \mathbf{R}^n$ is compact iff it is closed and bounded.

Complex rotations

In this section, we relax the assumption that we are working over $\mathbf{R}$. In mathematics (and physics) we often want to do this because the fundamental theorem of algebra is false over $\mathbf{R}$ (the field $\mathbf{R}$ is not algebraically closed).

Exercise 9. *Diagonalize the 2×2 rotation matrix from problem 1 above. Notice that you have to extend to complex-valued entries, that is, we must work over $\mathbf{C}$.*

Exercise 10. *Calculate the generator of this transformation.*

Exercise 11. *Describe the group $SL(2, \mathbf{C})$.*

When working over $\mathbf{C}$ we are often interested, not in the Kronecker inner product, but in a sesquilinear form called the Hermitian inner product. Explicitly, let u and v denote elements in $\mathbf{C}^n$ with complex components $u^i{}_{i=1}^n$ and $v^i{}_{i=1}^n$. Then the Hermitian inner product on $\mathbf{C}^2$ is defined to be

$$\langle u, v \rangle = \sum_{i=1}^{n} u^i v^i$$

The unitary group $U(n)$ is defined to be the subgroup of the general linear group $GL(\mathbf{C}^n)$ that preserves this Hermitian form.[11]

Exercise 12. *Describe the group $U(1)$ and its Lie algebra $\mathfrak{u}(1)$. Same for $SU(1)$ and $\mathfrak{su}(1)$.*

Exercise 13. *Show that the diagonal rotation matrix you found above is unitary.*

Exercise 14. *Starting over, parameterize a generic element of $SU(2)$ by starting with $GL(2, \mathbf{C})$, imposing $\det = 1$ to reduce to $SL(2, \mathbf{C})$, and finally imposing unitarity. What is the real dimension of this group? Prove that if you forget about the group structure, this space is a 3-dimensional sphere. In other words, show that $SU(2) \simeq S^3$.*

Exercise 15. *Using the exponential map and the fact that unitary operators are normal, describe the Lie algebras $\mathfrak{u}(n)$ and $\mathfrak{su}(n)$.*

Next, we define the Pauli matrices

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad , \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad , \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These are Hermitian and traceless. Your generator from problem 10 is anti-hermitian and traceless and related to the Pauli matrices by multiplication by $\pm i$.

Exercise 16. *Calculate the commutators of the Pauli matrices.*

1. *What do you notice?*

[11] It is perfectly fine to define/study $O(\mathbf{C}^n)$ and $SO(\mathbf{C}^n)$, but usually the relevant structure in the complex setting is not the orthogonal one.

2. *Define the generators $S_i = -\frac{i}{2}\sigma^i$, and write down the commutation relations of these.*
3. *Do you stand by your answer to problem 8? Comment in detail about what these calculations show.*

Exercise 17.

1. *Calculate $e^{\alpha S_1}$, $e^{\beta S_2}$, and $e^{\gamma S_3}$.*
2. *How do the ranges of these "angles" compare to those of the Euler angles?*
3. *How many elements of $SU(2)$ are there for each element of $SO(3)$? (This question is not about generators or dimensions since both groups have real dimension 3. Rather, if $R(\alpha,\beta,\gamma) \in SO(3)$ is some rotation, to which element(s) $S(\alpha,\beta,\gamma) \in SU(2)$ does it correspond?*

Exercise 18. *Starting from the Pauli matrices, find a basis for $\mathfrak{sl}(2,\mathbf{R})$. Find one for $\mathfrak{gl}(2,\mathbf{R})$.*

§Group representations The standard to study groups is to look at how they act on various things. For example, rotations rotate vectors in a very specific way, as demonstrated explicitly in the 3×3 matrix representation. But we can make larger matrices that satisfy the same algebra, and we can even make them infinite-dimensional.

Exercise 19. *Define generators $L_{ij} : \mathcal{C}^\infty(\mathbf{R}^3) \to \mathcal{C}^\infty(\mathbf{R}^3)$ acting on functions by*

$$L_{ij} f = x_i \partial_j f - x_j \partial_i f \qquad \forall f \in \mathcal{C}^\infty(\mathbf{R}^3)$$

Here, we are lowering the index $x_i = \delta_{ij} x^j$ of the coordinates $x^i{}_{i=1}^3$ of $\mathbf{R}^3$. Show that they generate rotations by calculating and comparing their commutators. What is the real dimension of this representation?

In addition, we saw that there was a 2×2 generalization of the algebra of rotations. It also acted on things that are not vectors. These "spinors" are elements of a 2-complex-dimensional space.

§Spin groups Above, you found that the $SU(2)$ rotations corresponding to opposite points on S^3 represented the same element in $SO(3)$. We now study this correspondence in more detail.

First, let us make precise the isomorphism $\mathfrak{su}(2) \simeq \mathbf{R}^3$. To do this, we want to associate to every vector in $\mathbf{R}^3$ an anti-hermitian 2×2 matrix. However, we already have a basis for these given by $S_i{}_{i=1}^3$: For every vector, we associate a matrix by contracting with this basis: where summation is implied.[12]

Exercise 20. *Explicitly write out the matrix v defined above.*

Exercise 21. *Use the trace to define the inverse map $\mathfrak{su}(2) \ni v \mapsto \mathbf{v} \in \mathbf{R}^3$.*

[12] In physics, this map is referred to as the Feynman slash because he used a slash through a (co-)vector to denote the matrix version of it.

A group has two natural actions on itself called the left and right action: Let $h \in G$ and define the left-multiplication $L_h : G \to G$ by $L_h(g) = hg$. Similarly, define the right action $R_h(g) = gh$.

Exercise 22. *Show that the left and right actions commute.*

Exercise 23. *...*